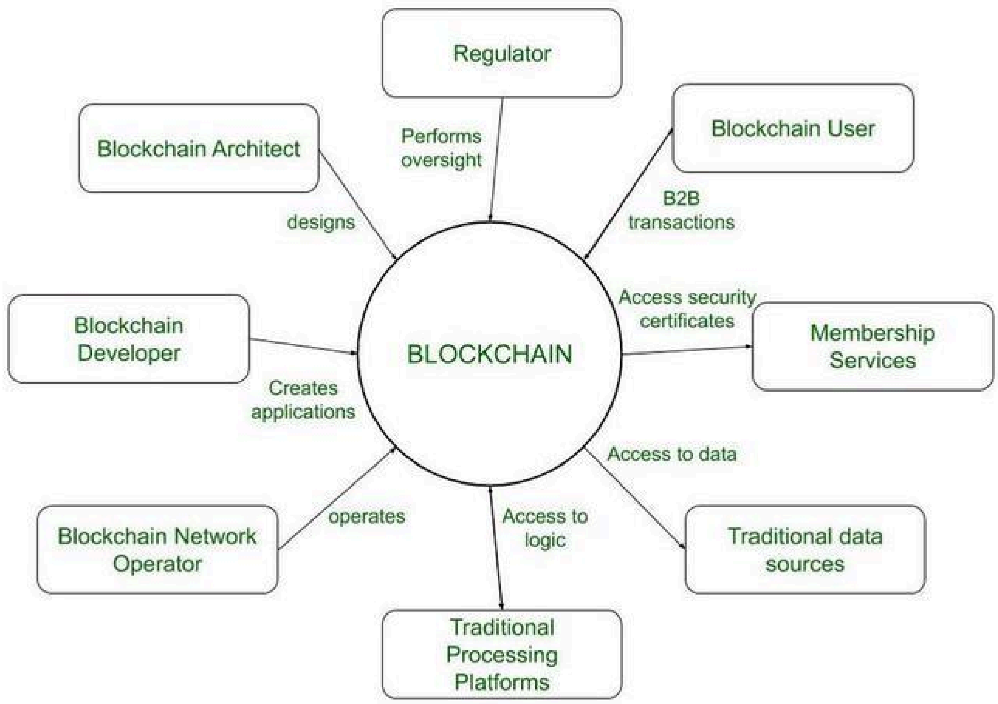
Vivekanand Education Society's Institute of Technology, Chembur, Mumbai
Department Of Computer Engineering
Academic Year : 2025-26
MID TERM TEST

Class : D17	**Division: A / B / C**
Semester: VII	**Subject: Blockchain**
Date: 11/09/2025	**Time: 10.30 am - 11.30 am**

Q.1 a) Enlist the various actors and mention their role in the Blockchain Ecosystem. (2M)

Actors:

1. **Users/Clients** – Initiate transactions.
2. **Nodes** – Maintain a copy of the blockchain and validate transactions.
3. **Miners/Validators** – Verify transactions, solve puzzles, and add blocks.
4. **Developers** – Build and maintain blockchain protocols/applications.
5. **Regulators/Governments** – Frame policies and ensure compliance.



Q. 1 b) **What is a Golden nonce? How is it calculated? (2M)**

- **Golden Nonce**: A specific value of the nonce that, when combined with block data and hashed, produces a hash lower than the target difficulty.
- **Calculation**: Miners keep incrementing the nonce and hashing the block header until the output hash meets the target condition. The nonce that satisfies it is the *Golden Nonce*.

Q. 1 c) What is meant by mining difficulty? How is it computed? (2M)

- **Mining Difficulty:** A measure of how hard it is to find a hash below the target value.
- **Computation:** Adjusted every 2016 blocks in Bitcoin →

$$\text{Difficulty} = \frac{\text{Target Time}}{\text{Actual Time}}$$

If blocks are mined too fast → difficulty increases, if too slow → decreases.

Q. 1 d) Enlist the characteristics of a Blockchain. (2M)

1. **Decentralized** – No central authority.
2. **Immutability** – Once recorded, data cannot be altered.
3. **Transparency** – All participants can view the ledger.
4. **Security** – Cryptography ensures integrity.
5. **Consensus-driven** – Decisions via mechanisms like PoW, PoS.

Q. 1 e) How does a miner select the transactions to be added into a block? (2M)

- Miners select transactions from the **mempool**.
- Preference given to:
    1. **Higher transaction fees** (maximize reward).
    2. **Valid transactions only** (digitally signed, no double-spending).
    3. **Block size limit** (e.g., 1 MB in Bitcoin).

Q. 1 f) What is an orphaned Block? (2M)

- A block that was mined but **not included in the main chain**.
- Occurs when two miners produce valid blocks at the same time → network selects the longer chain, leaving the other block orphaned.

Q. 2 a) Differentiate between Hot and Cold Wallets. (5M)

Feature	Hot Wallet	Cold Wallet
Connectivity	Connected to the Internet	Offline (not connected)
Security	Less secure, prone to hacks	Highly secure against online attacks
Convenience	Easily accessible for frequent transactions	Less convenient, suitable for storage
Examples	Mobile apps, web wallets	Hardware wallets, paper wallets
Usage	Daily trading and payments	Long-term holding of crypto assets

Q.2 b) Explain with an example how the wallet balance is computed in Blockchain. (5M)

- Wallets do not store coins; they store **private/public keys**.
- Balance = sum of **unspent transaction outputs (UTXOs)** associated with wallet's public address.

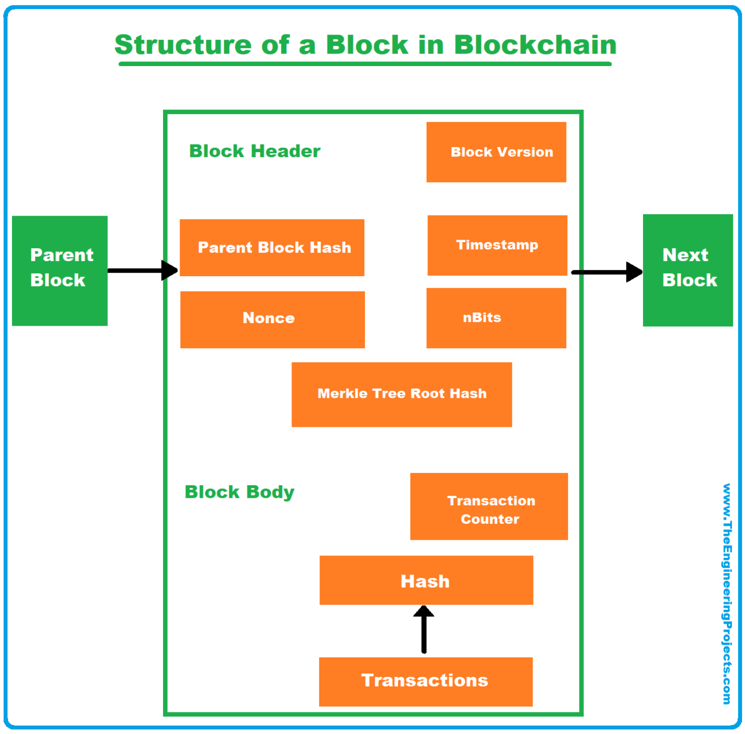
Example:

- UTXOs: 2 BTC, 3 BTC, 1 BTC → Total Wallet Balance = 6 BTC.
- If user spends 4 BTC →
    - 3 BTC UTXO consumed + 1 BTC from another → transaction created.
    - Remaining 2 BTC sent back as "change" → new UTXO for the wallet.

Q. 3 a) With the help of a neat diagram explain the basic structure of a Block in Blockchain. (5M)

Block Structure:

1. **Block Header**
    - Block number
    - Previous block hash
    - Timestamp
    - Nonce
    - Merkle root
2. **Block Body :** List of valid transactions



Q. 3 b) With the help of a neat diagram explain the process of mining in Bitcoin. (5M)

Steps in Mining:

1. Transactions collected in mempool.
2. Miner selects transactions and forms block.
3. Block header is hashed with varying nonce.
4. If hash < target → block is valid.
5. Block broadcasted, added to blockchain, miner rewarded.